23 A circular loop of wire with radius of 0.20 m is placed in a uniform magnetic field of 1.2 T that is normal to the plane of the loop. The loop is rotated about an axis along a diameter at a rate of 50 revolutions per minute. What is the maximum instantaneous e.m.f. induced?

A 0.13 V **B** 0.15 V **C** 0.79 V **D** 47 V

$$\varepsilon = -\frac{d\phi}{dt} = -\frac{d(BA\cos\omega t)}{dt} = \omega BA\sin\omega t$$

$$\varepsilon_{\text{max}} = \omega BA = \frac{50x2\pi}{60}x1.2x\pi x0.20^2 = 0.79V$$

24 A small circular coil lies inside a large coil. The two coils are horizontal and concentric. The large